Selective repeat

sender

data from above:

- if available seq # in window, send pkt

timeout(n):

- resend pkt n, restart timer

ACK(n) in [sendbase,sendbase+N]:

- mark pkt n as received
- if n oldest unACKed pkt, advance window base to next unACKed seq #
- restart timer for next unACKed seq #

receiver

pkt n in [rcvbase, rcvbase+N-1]

- send ACK(n)
- out-of-order: buffer
- in-order: delivery (like GBN, deliver buffered, in-order pkts), advance window to next not-yet-received pkt

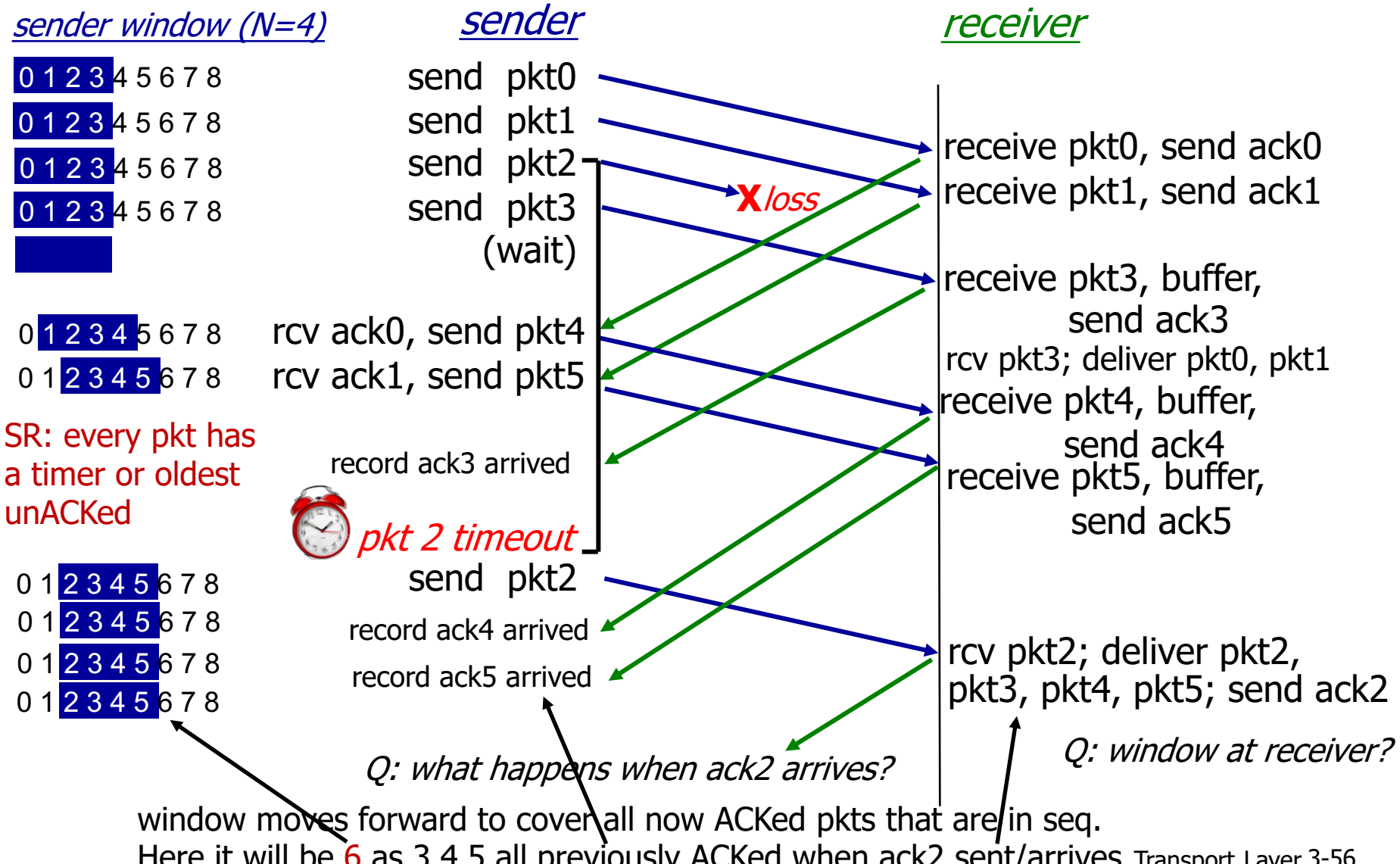
pkt n in [rcvbase-N,rcvbase-1]

- ACK(n) (ACK lost, repeat)

otherwise:

- ignore

Selective repeat in action



Selective repeat: dilemma

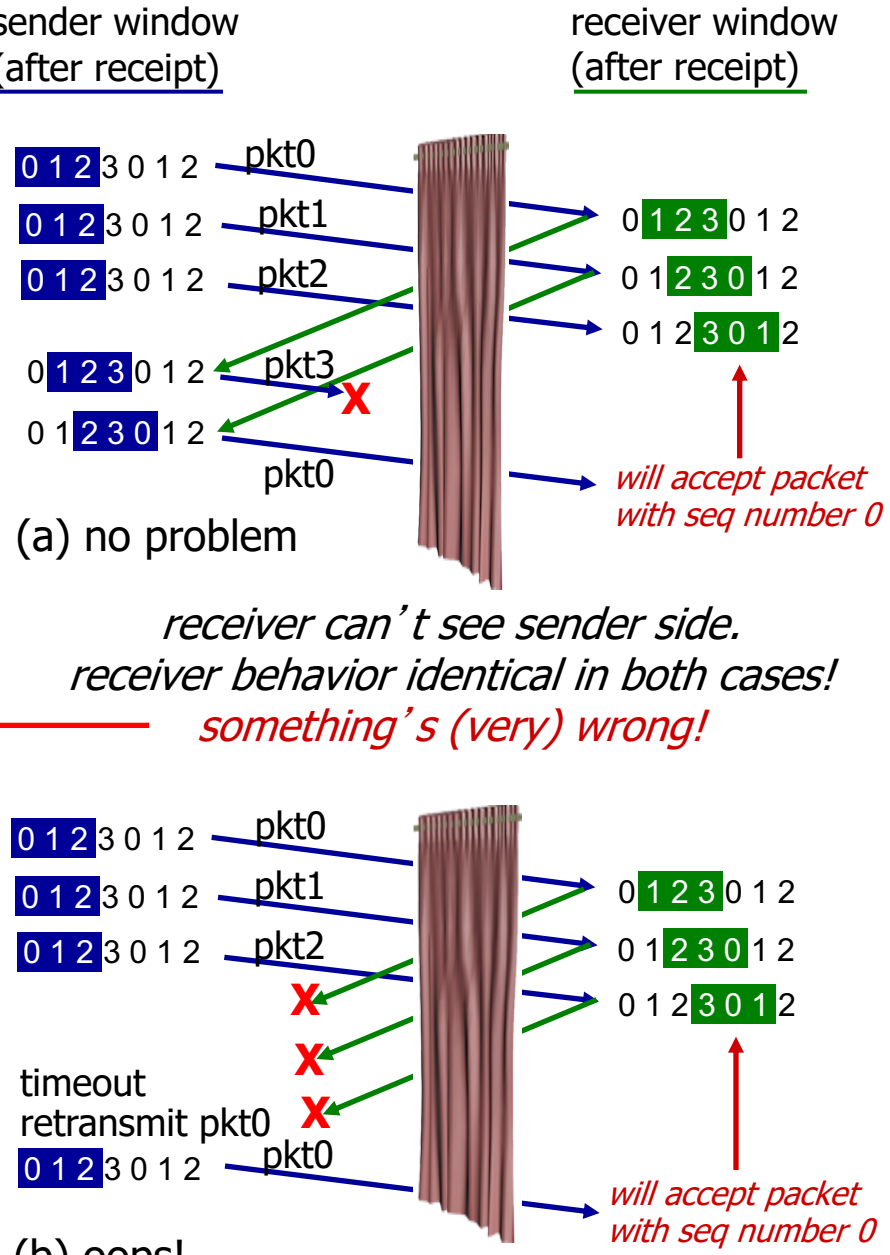
example:

- seq #'s: 0, 1, 2, 3
- window size=3

- receiver sees no difference in two scenarios!
- duplicate data accepted as new in (b)

Q: what relationship between seq # size and window size to avoid problem in (b)?

A: window size must be << than seq # space → half or less



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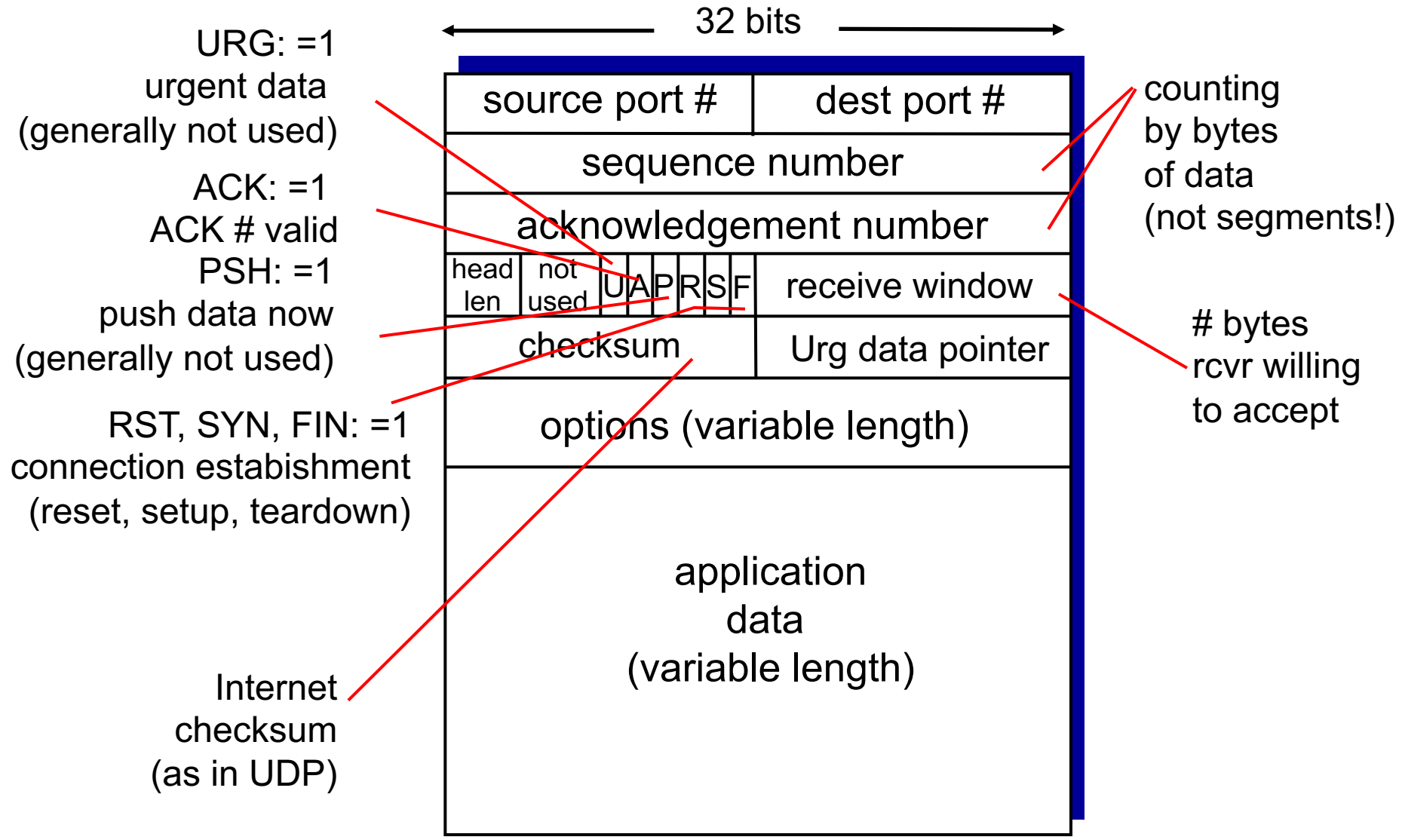
TCP: Overview RFCs: 793,1122,1323, 2018, 2581

- point-to-point:
  - one sender, one receiver
- reliable, in-order *byte stream:*
  - no "message boundaries"
- pipelined:
  - TCP congestion and flow control set window size
- full duplex data:
  - bi-directional data flow in same connection
  - MSS: maximum segment size
- connection-oriented:
  - handshaking (exchange of control msgs) inits sender, receiver state before data exchange
- flow controlled:
  - sender will not overwhelm receiver

TCP segment structure

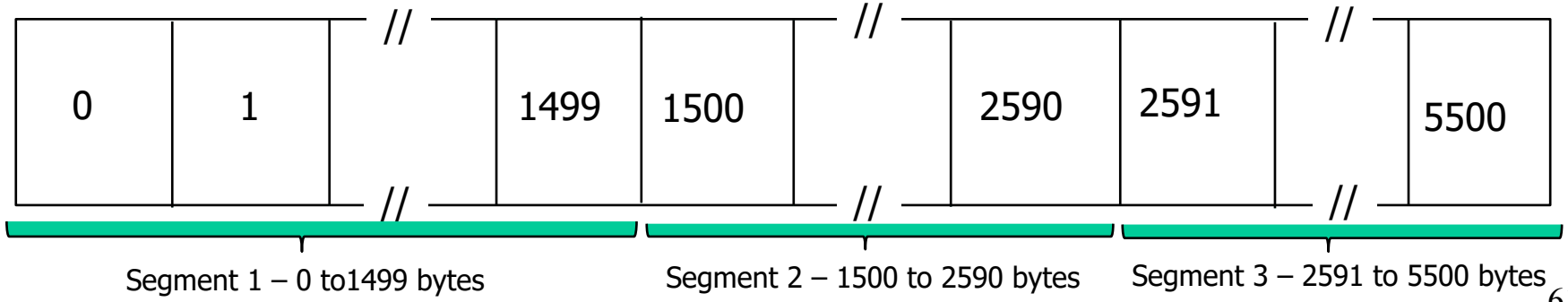


Sequence Nos

- sequence number is 32 bits long
  - the range of SeqNo is

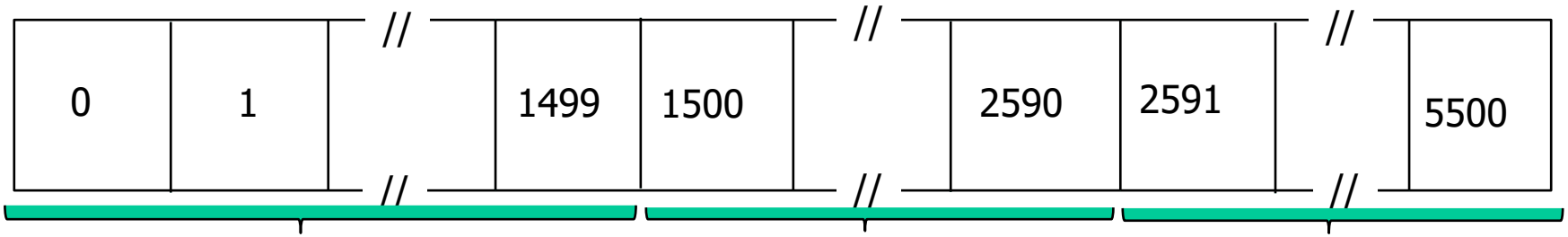
$$0 <= \text{SeqNo} <= 2^{32} - 1 \approx 4.3 \text{ Gbyte}$$

  - a sequence number identifies a specific byte in the byte stream. Each byte has a sequence number
  - an Initial Sequence Number (ISN) for a new connection is picked randomly at each end of the connection and is exchanged during connection establishment



Acknowledgement Nos

- ACKs can be piggybacked
  - a segment from A -> B can contain an acknowledgement for data sent in the B -> A direction
- a hosts uses the acknowledgment field to ACK a pkt
  - if a host sends an ACK# in a segment it sets the "**ACK flag**"
- the ACK# contains the **next** Seq# that a receiving host is expecting to receive from the sender.
  - the ACK# for a segment with Seq# 0 and data length of 1500 bytes (0-1499) is = 1500
  - next byte TCP receiver is expecting is byte # 1500 (received 0-1499)
  - next segment sent from sender should have Seq# = 1500.



TCP seq. numbers, ACKs

sequence numbers:

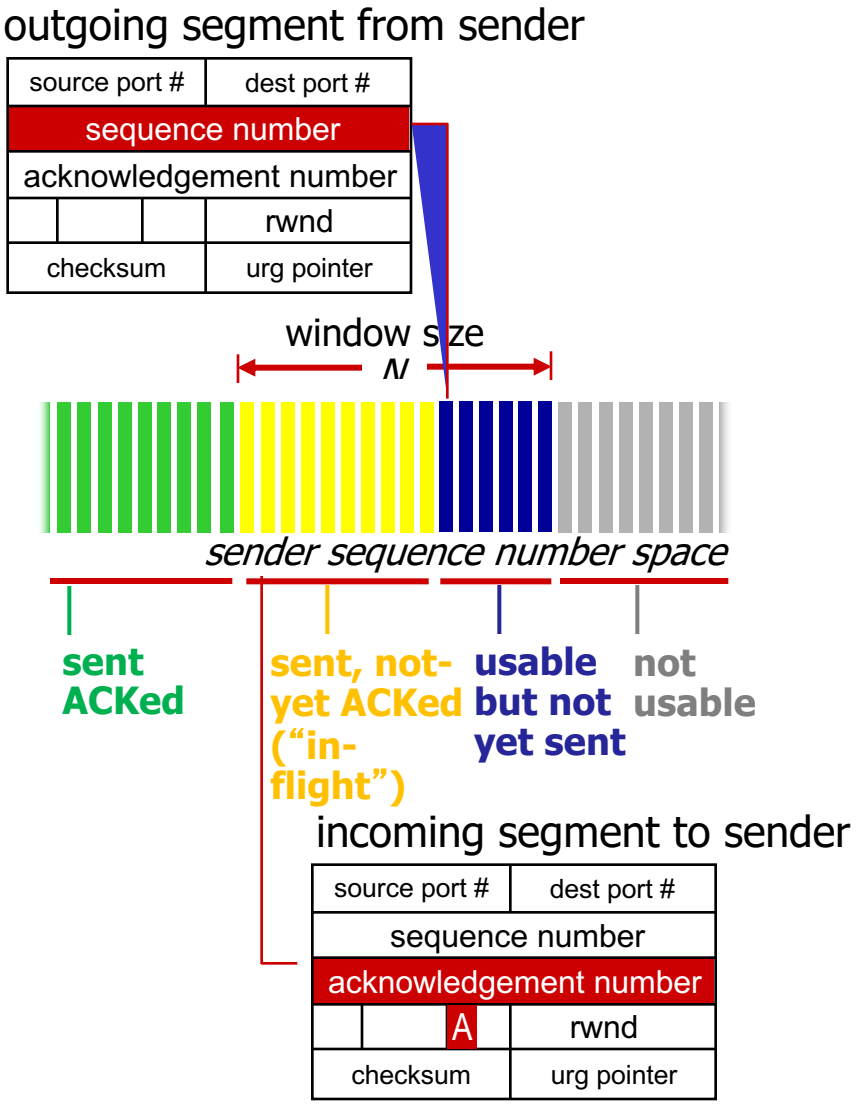
- byte stream "number" of first byte in segment's data

acknowledgements:

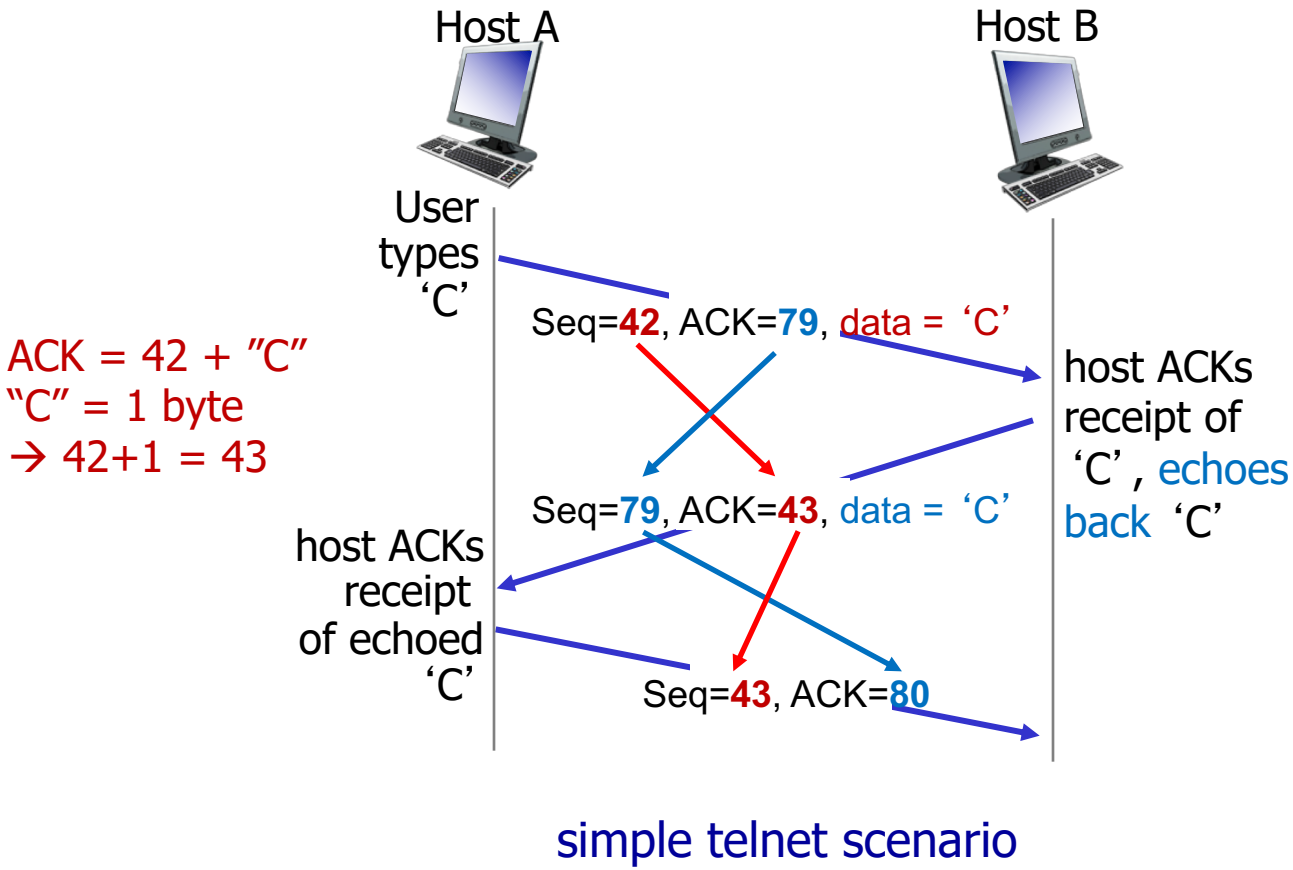
- seq # of next byte expected from other side
- cumulative ACK

Q: how receiver handles out-of-order segments?

A: TCP spec doesn't say, - up to implementor



TCP seq. numbers, ACKs



simple telnet scenario

TCP round trip time, timeout

Q: how to set TCP timeout value?

- longer than RTT
  - but RTT varies
- *too short*: premature timeout, unnecessary retransmissions
- *too long*: slow reaction to segment loss

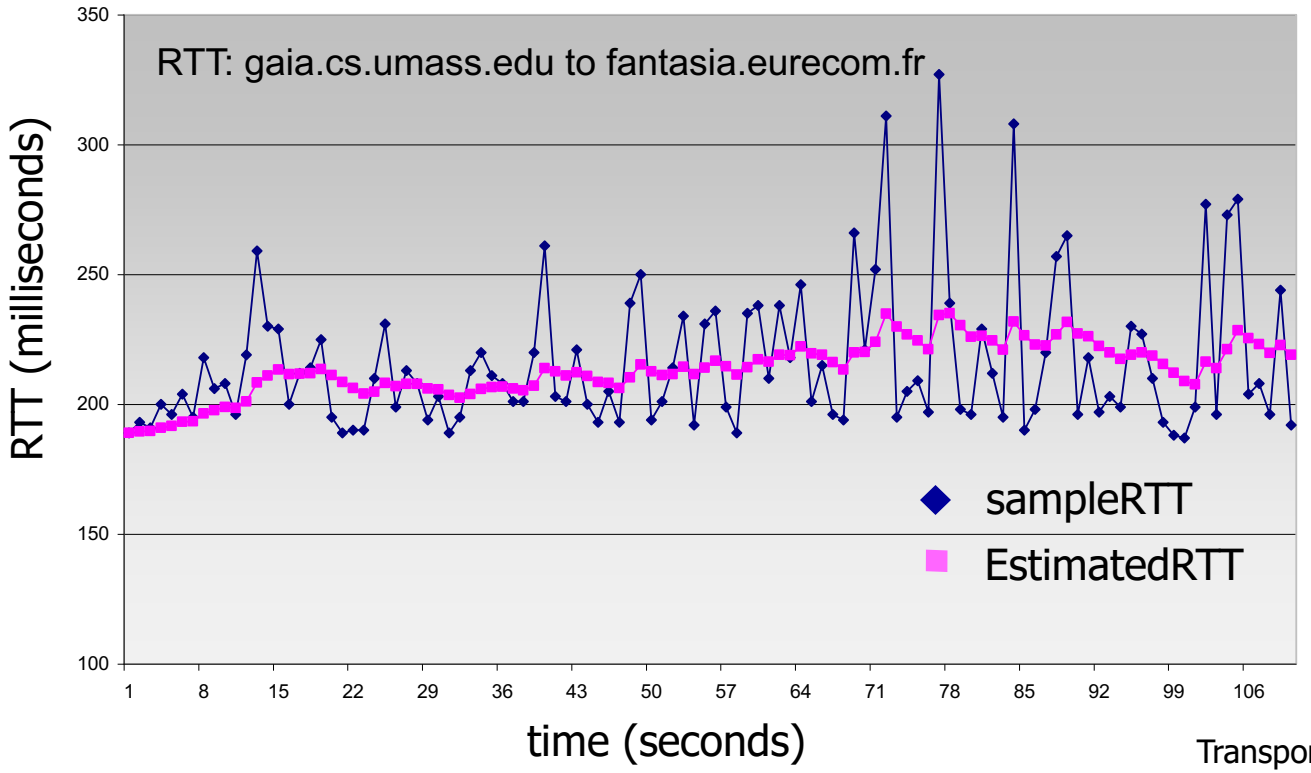
Q: how to estimate RTT?

- **`SampleRTT`**: measured time from segment transmission until ACK receipt
  - ignore retransmissions
- **`SampleRTT`** will vary, want estimated RTT "smoother"
  - average over several *recent* measurements, not just current **`SampleRTT`**

TCP round trip time, timeout

EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT

- exponential weighted moving average
- influence of past sample decreases exponentially fast
- typical value: α = 0.125



TCP round trip time, timeout

- timeout interval: **EstimatedRTT** plus "safety margin"
  - large variation in **EstimatedRTT** -> larger safety margin
- estimate SampleRTT deviation from EstimatedRTT:

DevRTT = (1-β)*DevRTT +
β*|SampleRTT-EstimatedRTT|

(typically, β = 0.25)

TimeoutInterval = EstimatedRTT + 4*DevRTT

estimated RTT

"safety margin"

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TCP reliable data transfer

- TCP creates a reliable service on top of IP's unreliable service
  - pipelined segments
  - cumulative acks
  - single retransmission timer
- retransmissions triggered by:
  - timeout events
  - duplicate acks

let's initially consider simplified TCP sender:

- ignore duplicate acks
- ignore flow control, congestion control

TCP sender events:

data rcvd from app:

- create segment with seq #
- seq # is byte-stream number of first data byte in segment
- start timer if not already running
  - think of timer as for oldest unacked segment
  - expiration interval: `TimeOutInterval`

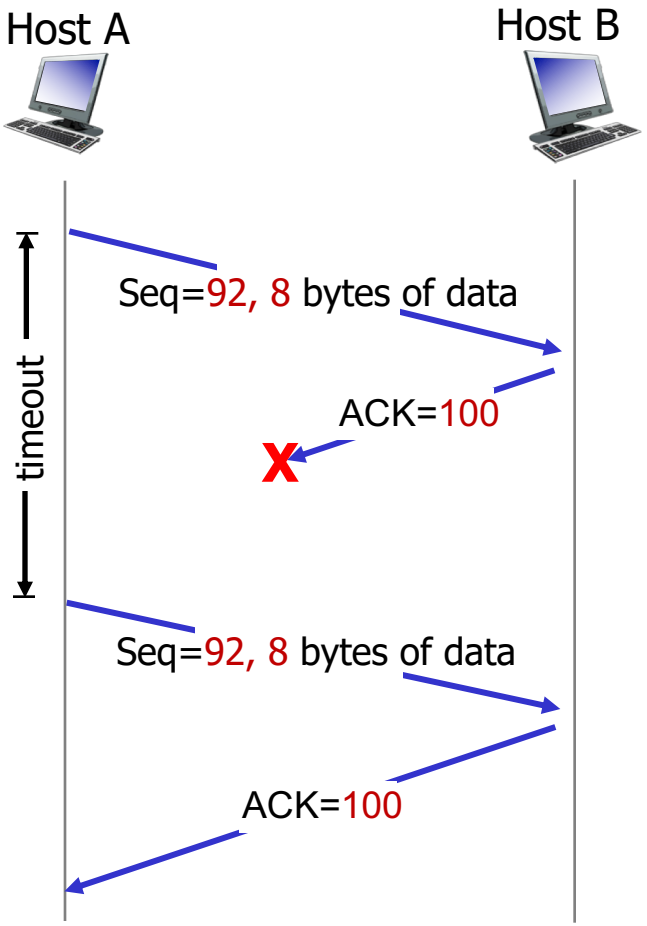
timeout:

- retransmit segment that caused timeout
- restart timer

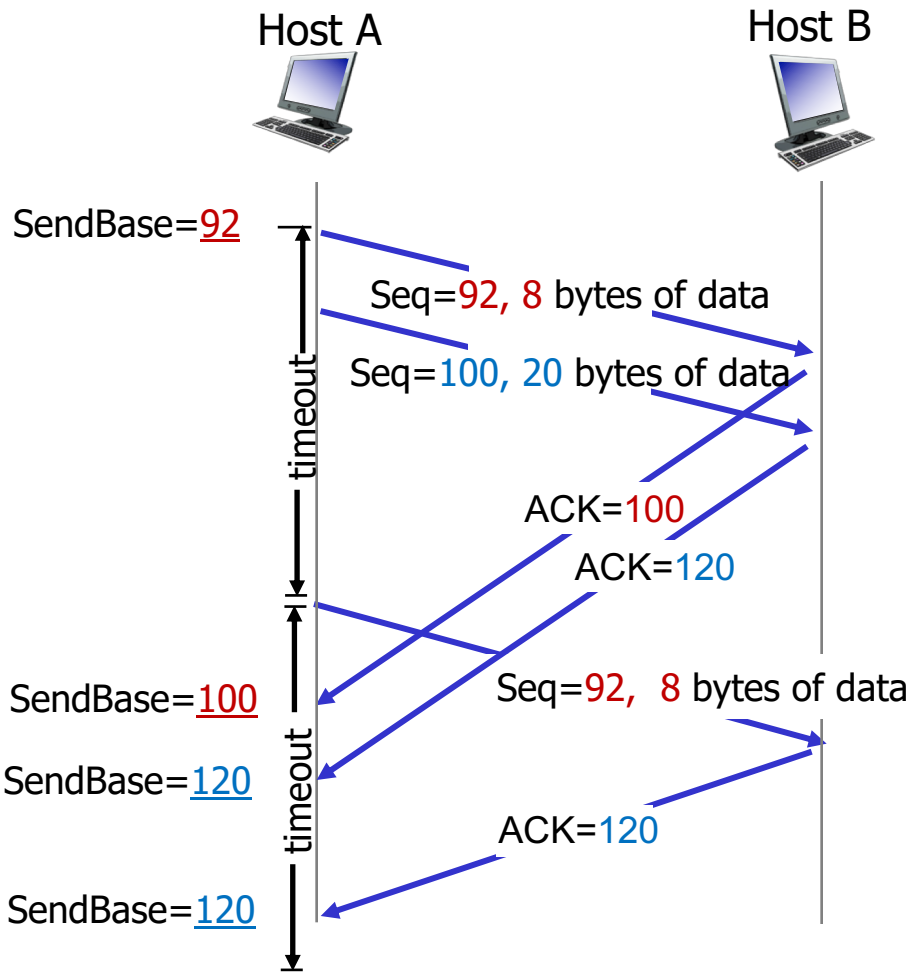
ack rcvd:

- if ack acknowledges previously unacked segments
  - update what is known to be ACKed
  - start timer if there are still unacked segments

TCP: retransmission scenarios

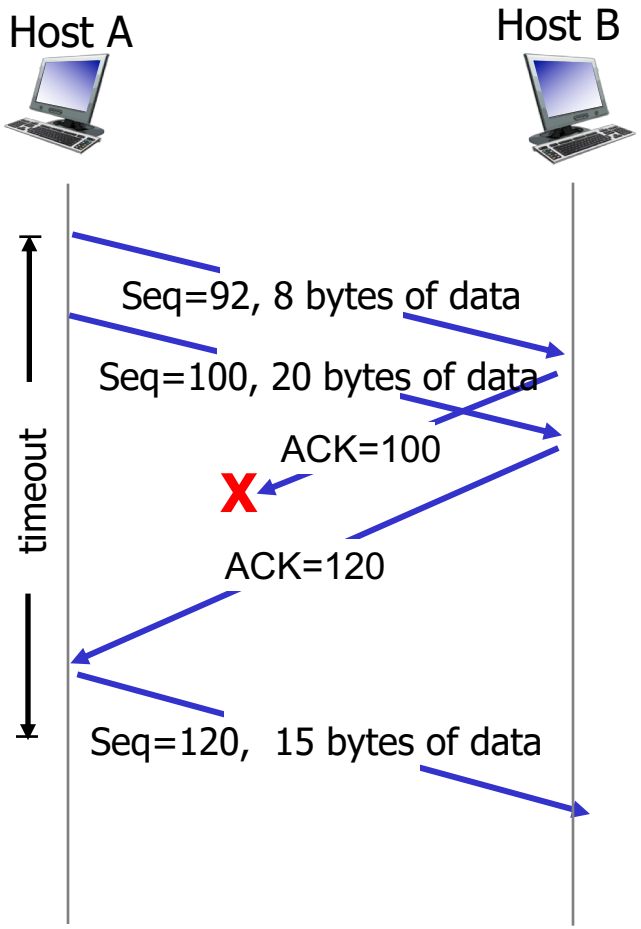


lost ACK scenario



premature timeout

TCP: retransmission scenarios



timeout better estimate, cumulative ACK

TCP ACK generation [RFC 1122, RFC 2581]

event at receiver	*TCP receiver action*
arrival of in-order segment with expected seq #. All data up to expected seq # already ACKed	delayed ACK. Wait up to 500ms for next segment. If no next segment, send ACK
arrival of in-order segment with expected seq #. One other segment has ACK pending	immediately send single cumulative ACK, ACKing both in-order segments
arrival of out-of-order segment higher-than-expect seq. # . Gap detected	immediately send *duplicate ACK*, indicating seq. # of next expected byte
arrival of segment that partially or completely fills gap	immediate send ACK, provided that segment starts at lower end of gap

TCP fast retransmit

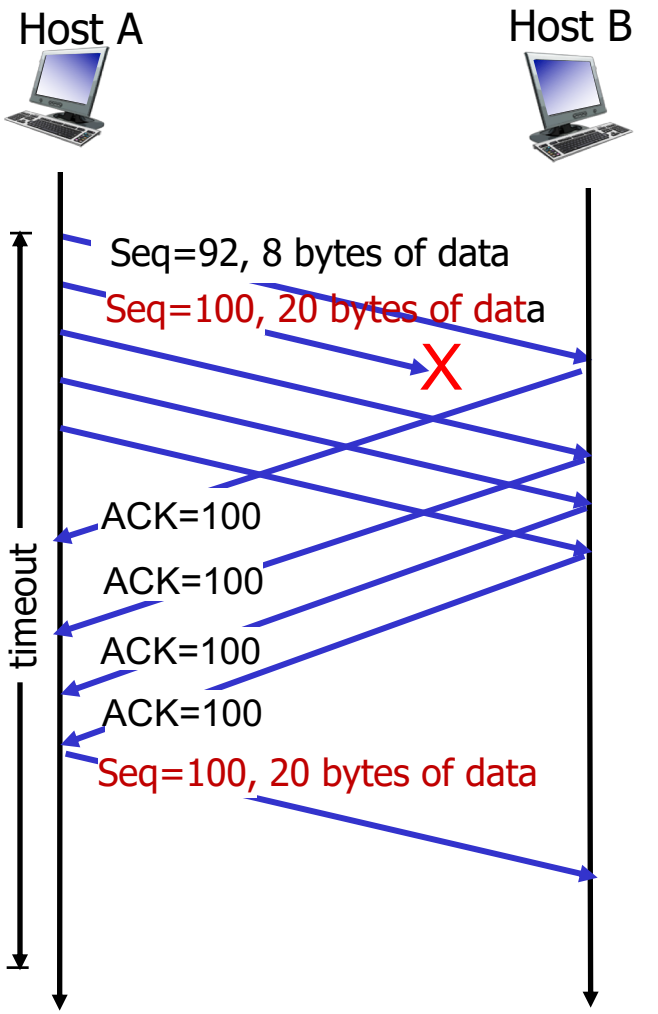
- time-out period often relatively long:
  - long delay before resending lost packet
- detect lost segments via duplicate ACKs.
  - sender often sends many segments back-to-back
  - if segment is lost, there will likely be many duplicate ACKs (every out of order segment will generate an ACK).

TCP fast retransmit

- if sender receives 1+3 ACKs for same data (i.e., original + "triple or 3" duplicate ACKs),
- resend unacked segment with smallest seq #
  - likely that unacked segment lost, so don't wait for timeout

Note: "triple" duplicate ACKs, means one ACK and 2 duplicates of that ACK

TCP fast retransmit



fast retransmit after sender receipt of triple duplicate ACK

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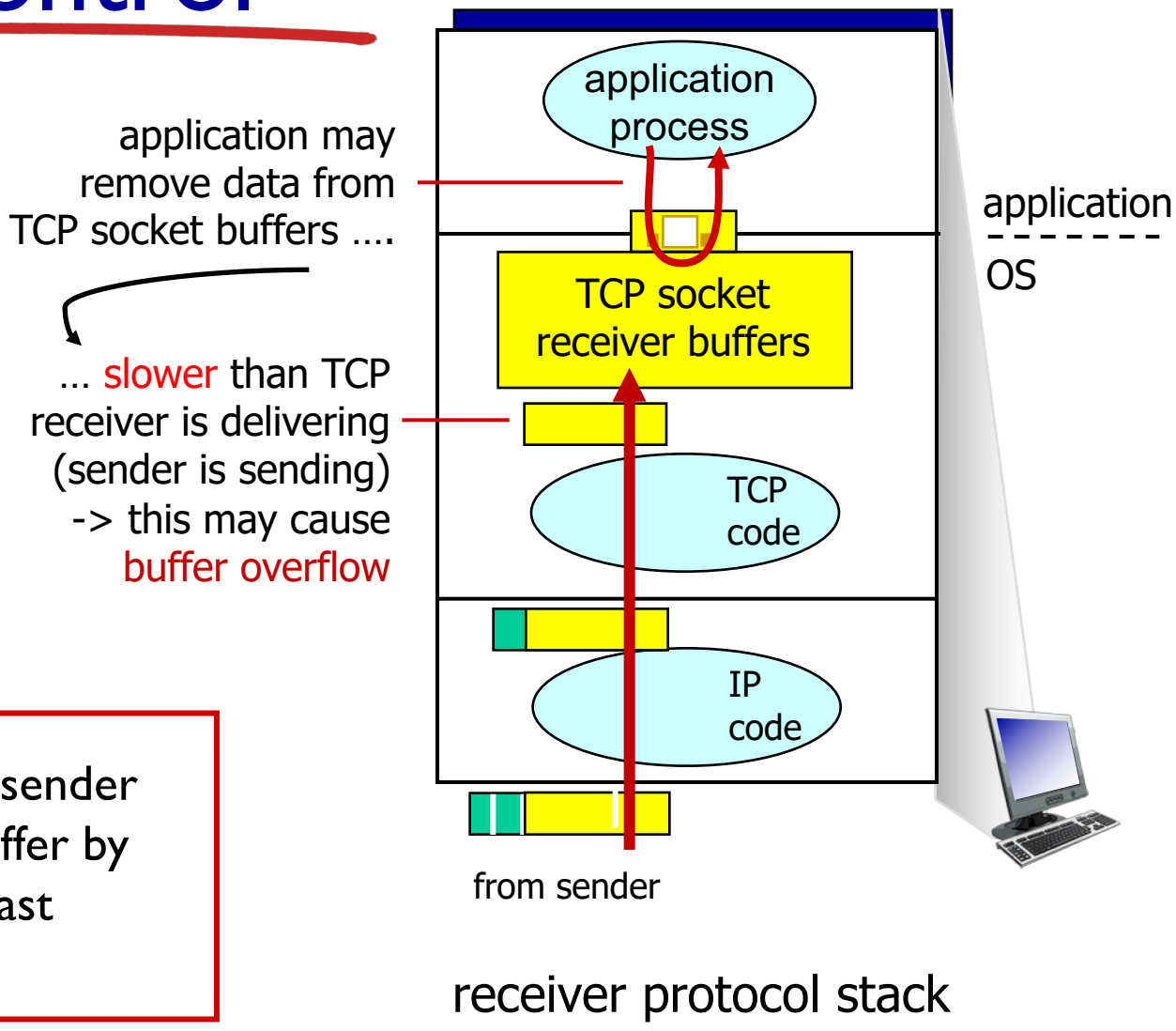
TCP flow control



receiver protocol stack

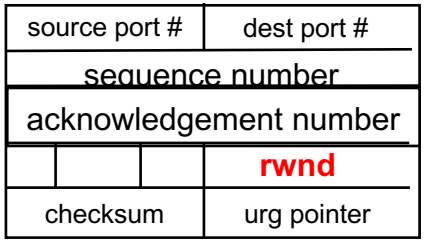
flow control

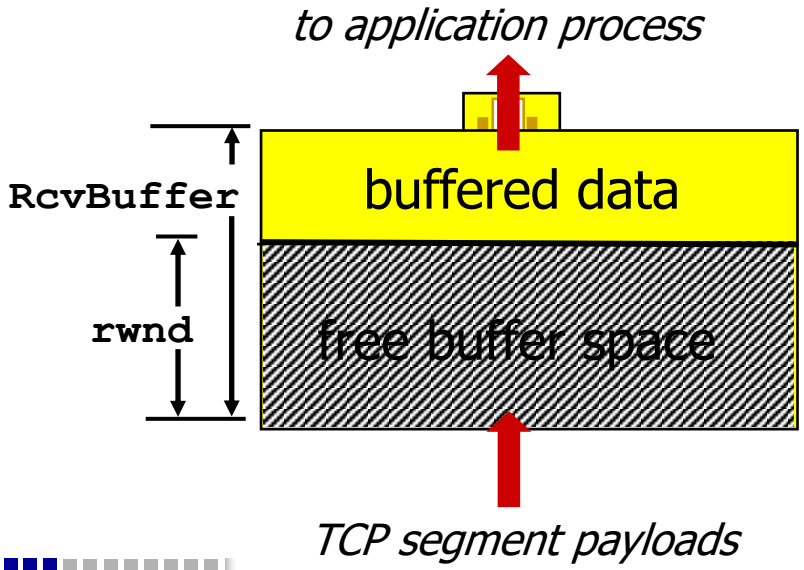
receiver controls sender, so sender won't overflow receiver's buffer by transmitting too much, too fast

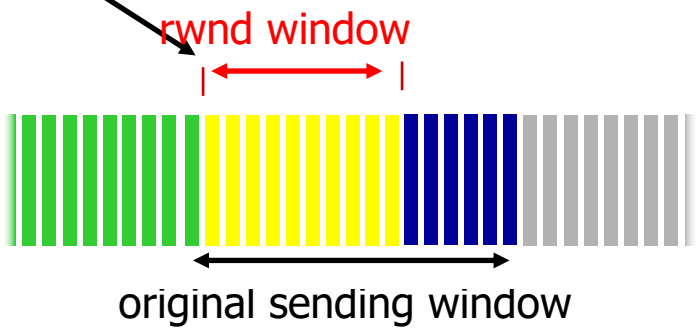
TCP flow control (FC)

- receiver "advertises" free buffer space by including **rwnd** value in TCP header of receiver-to-sender segments
  - **RcvBuffer** size set via socket options (typical default is 4096 bytes)
  - many operating systems auto adjust **RcvBuffer**
- sender limits amount of unacked ("in-flight") data to receiver's **rwnd** value (changes value of sending window → FC window = rwnd)
- guarantees receive buffer will not overflow

segment sent to sender from receiver







receiver-side buffering

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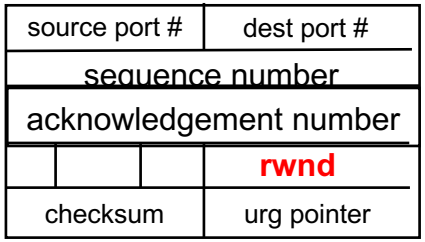
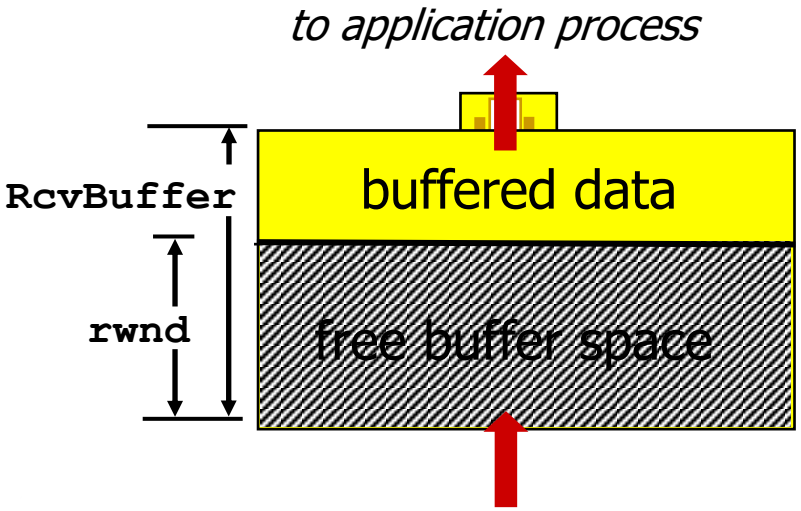
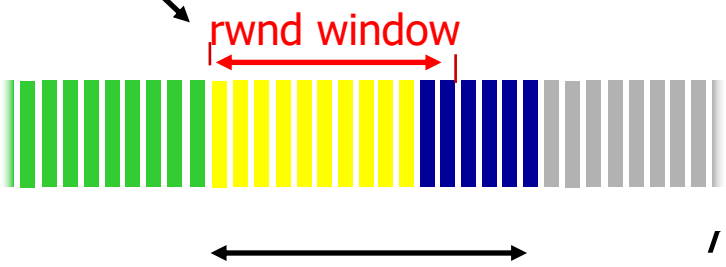


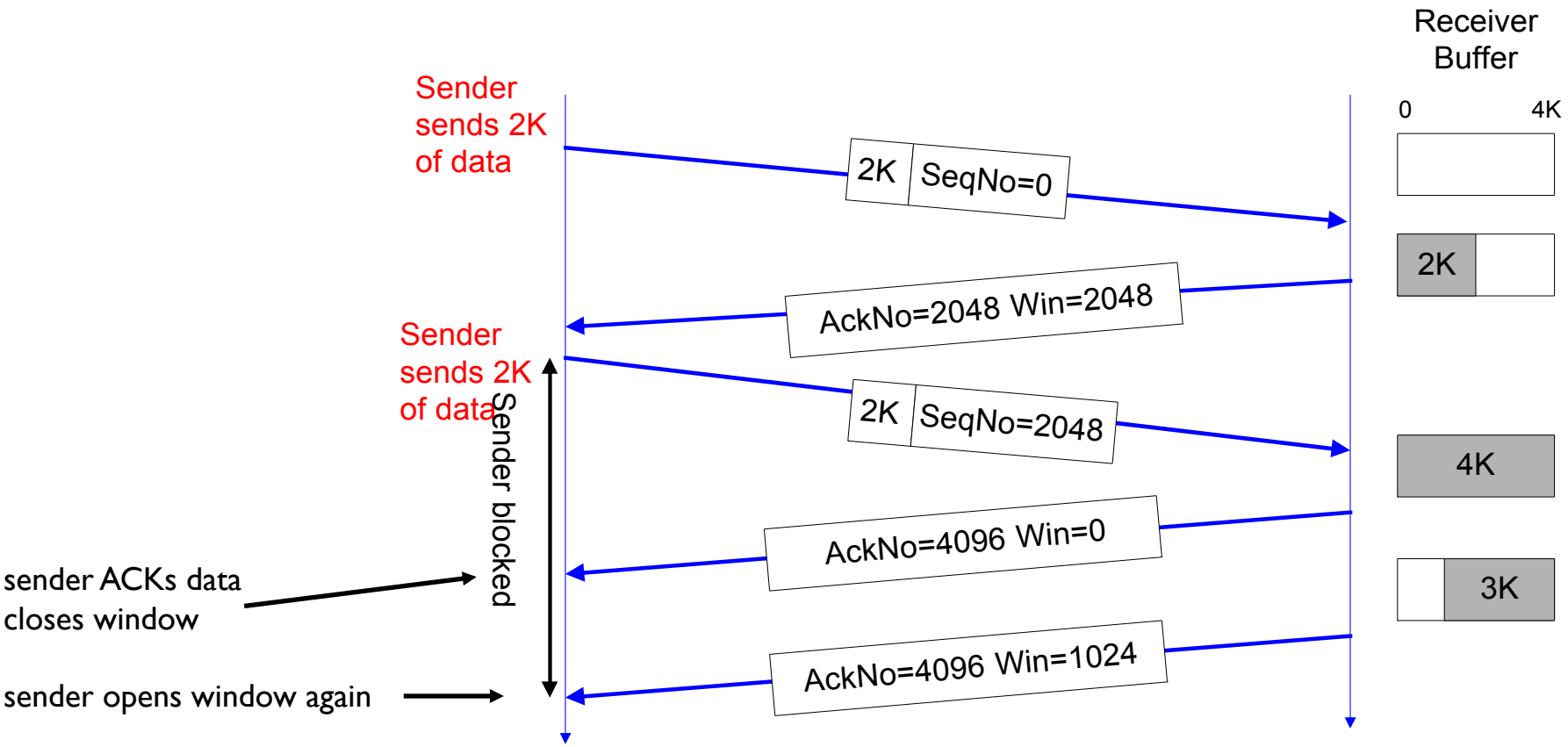




if rwnd > unACKed, then sender can send more packets up to rwnd

receiver-side buffering

Example


Receiver Buffer
0
4K
Sender sends 2K of data
2K
SeqNo=0
2K
AckNo=2048 Win=2048
Sender sends 2K of data
2K
SeqNo=2048
4K
Sender blocked
AckNo=4096 Win=0
3K
sender ACKs data closes window
AckNo=4096 Win=1024
sender opens window again

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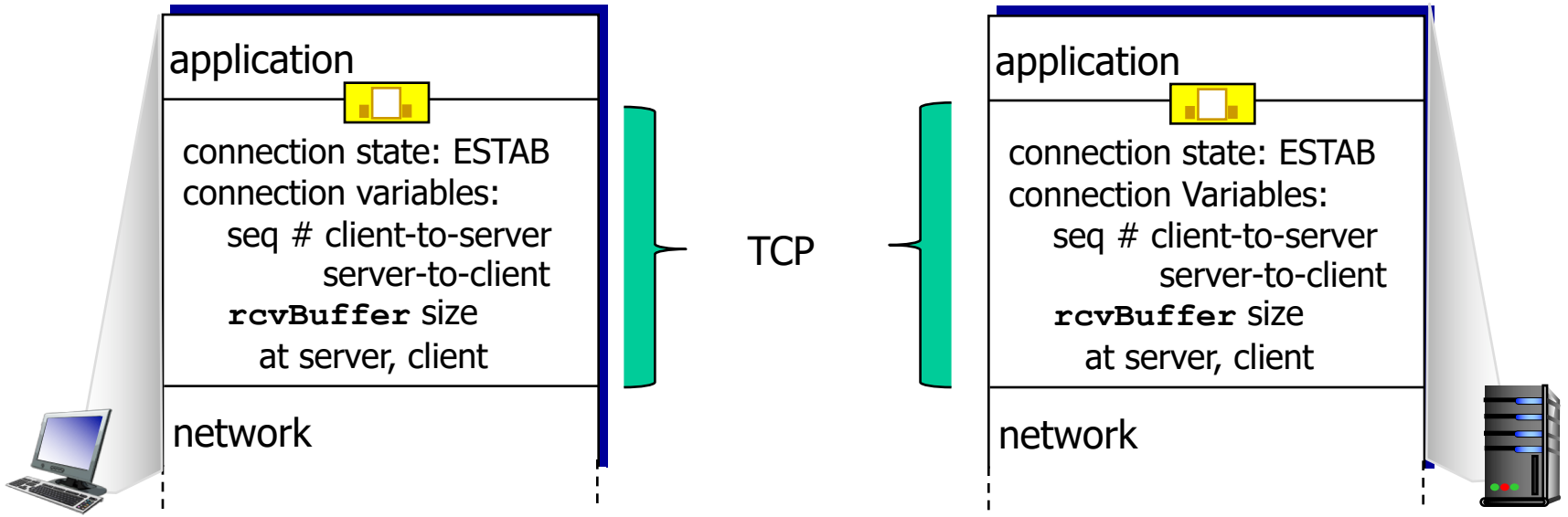
Connection Management

before exchanging data, sender/receiver "handshake":

- agree to establish connection (each knowing the other willing to establish connection)
- agree on connection parameters

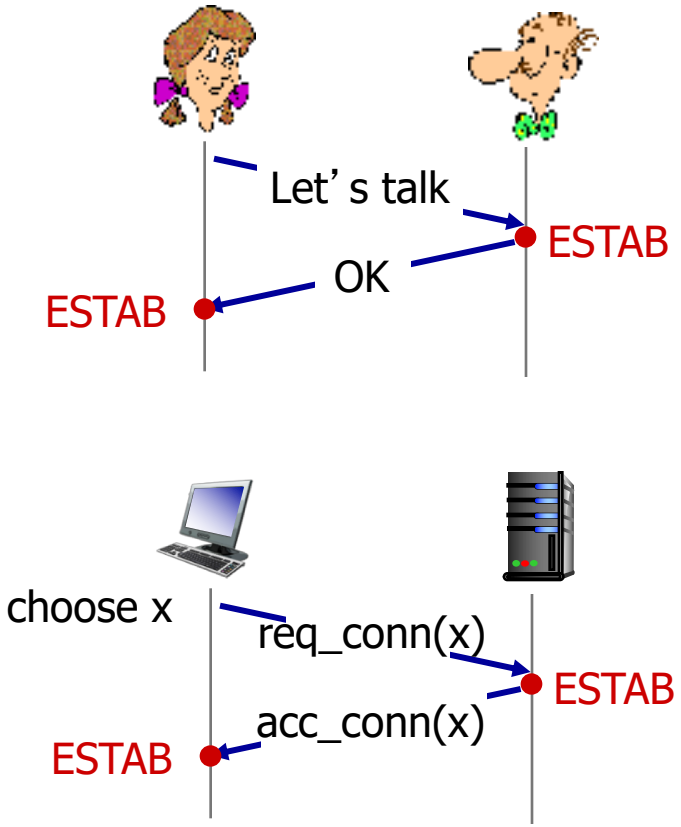



```
Socket clientSocket =
  newSocket("hostname","port
   number");
```

```
Socket connectionSocket =
   welcomeSocket.accept();
```

Agreeing to establish a connection

2-way handshake:

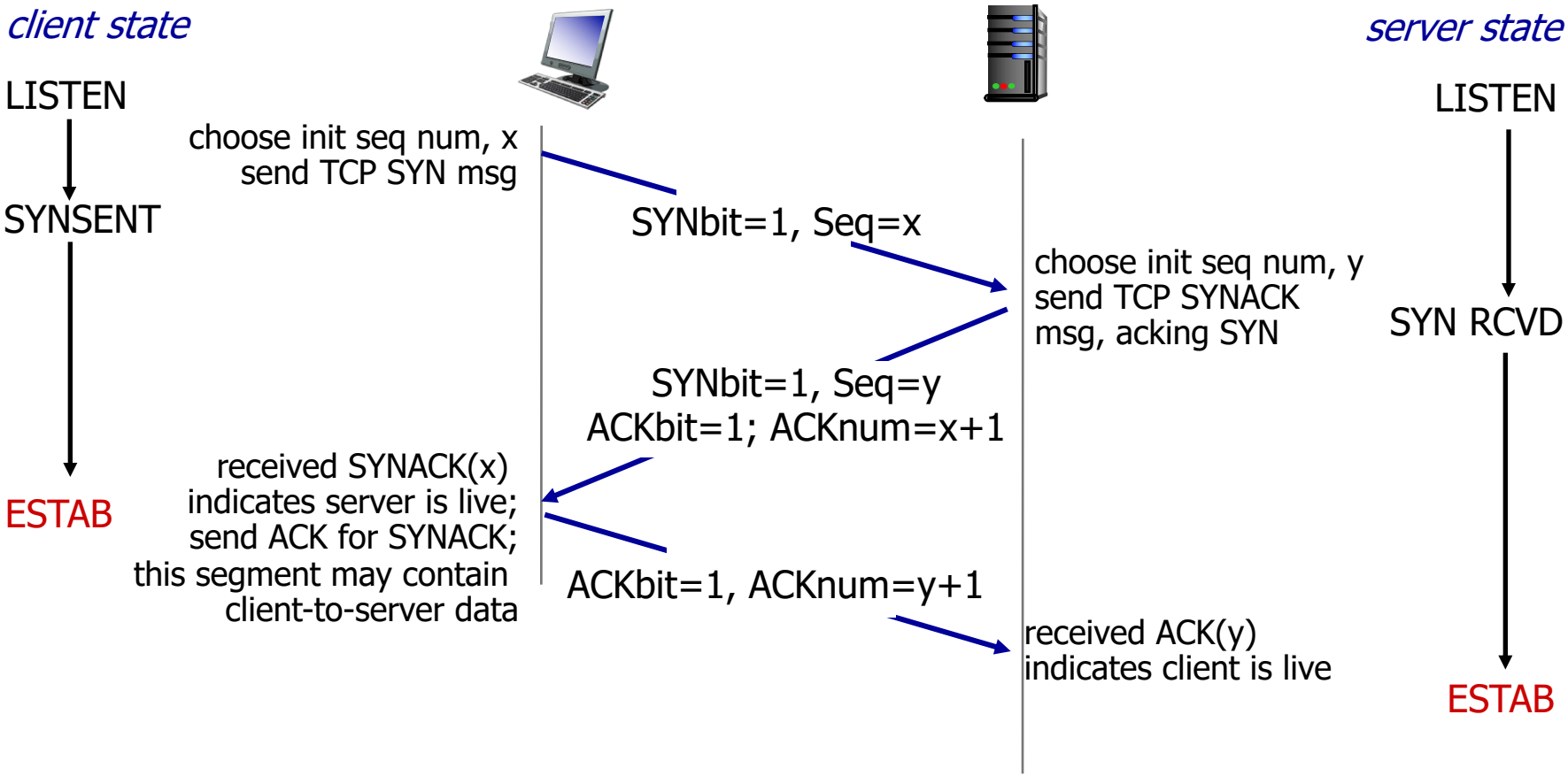


Q: will 2-way handshake always work in network?

A: No

- variable delays
- retransmitted messages (e.g. req_conn(x)) due to message loss
- message reordering
- can't "see" other side

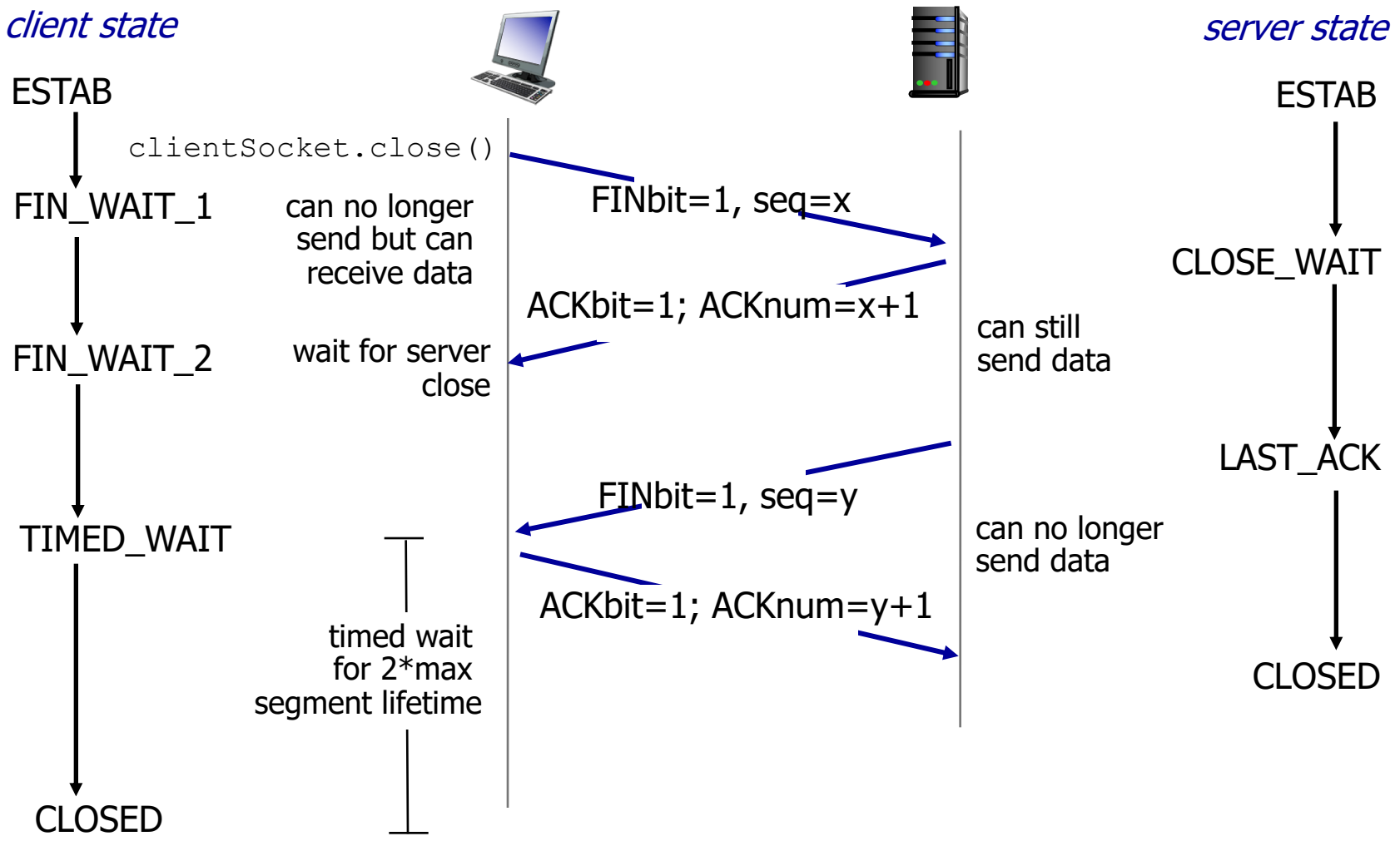
TCP 3-way handshake



TCP: closing a connection

- client, server each close their side of connection
  - send TCP segment with FIN bit = 1
- respond to received FIN with ACK
  - on receiving FIN, ACK can be combined with own FIN
- simultaneous FIN exchanges can be handled

TCP: closing a connection



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Principles of congestion control

congestion:

- informally: "too many sources sending too much data too fast for *network* to handle"
- different from flow control!
- manifestations:
  - lost packets (buffer overflow at routers)
  - long delays (queueing in router buffers)

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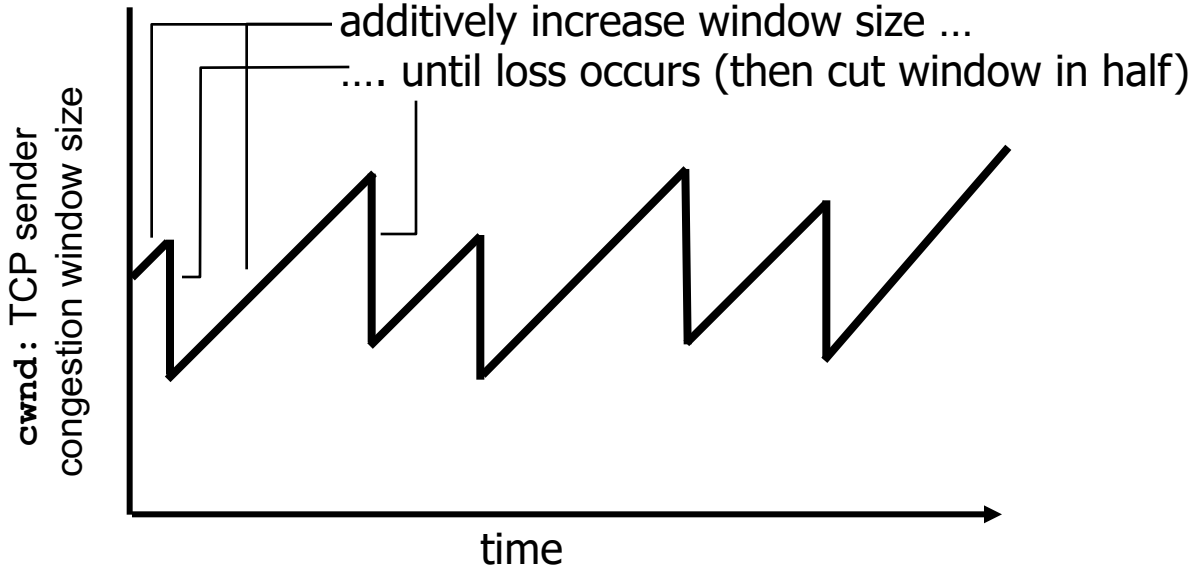
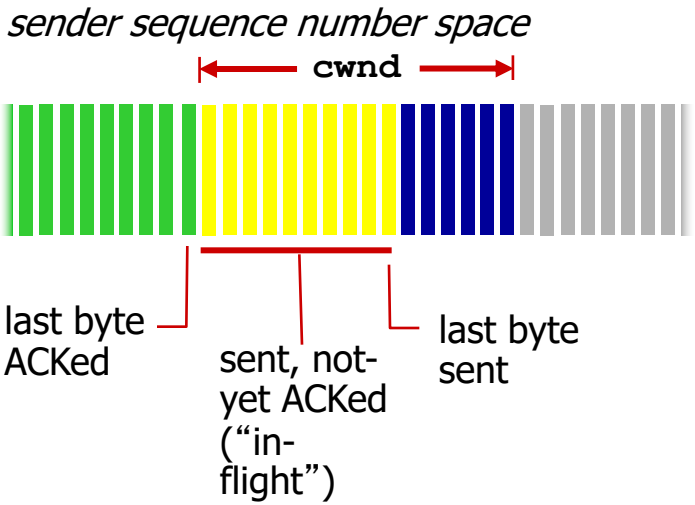
TCP congestion control: principles

- *approach:* sender increases transmission rate (window size), probing for usable bandwidth, until loss occurs
  - *Additive Increase:* increase **`cwnd`** by 1 MSS every RTT (i.e., received ACK) until loss detected
  - *Multiplicative Decrease:* cut **`cwnd`** in half after loss

AIMD saw tooth behavior: probing for bandwidth



TCP Congestion Control: details



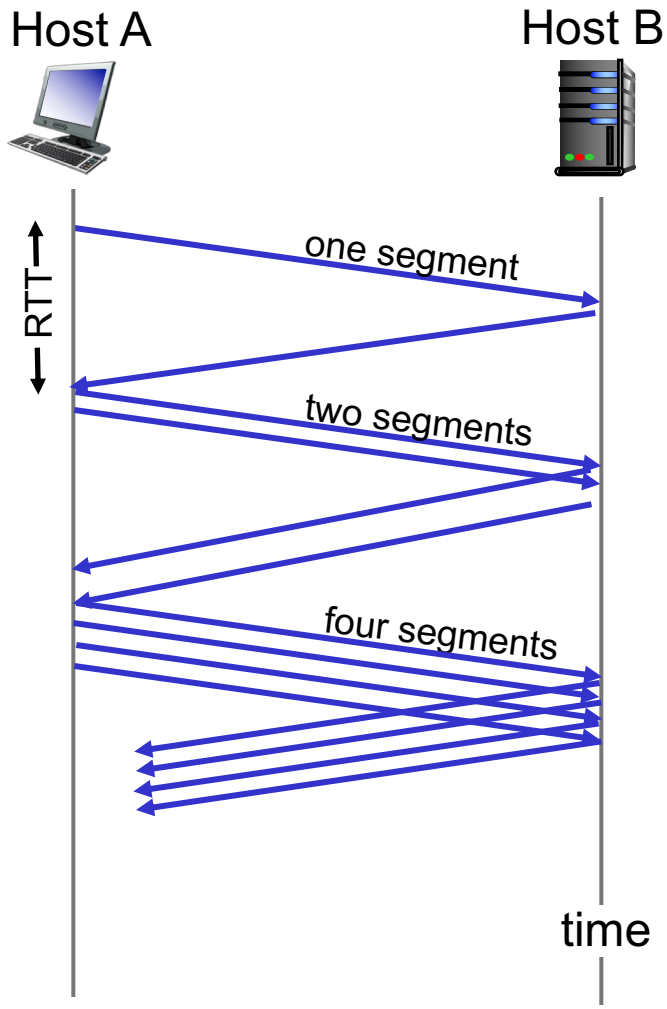
- sender limits transmission:

$$\texttt{LastByteSent} - \texttt{LastByteAcked} \leq \texttt{cwnd}$$

- **cwnd** is dynamic, function of perceived network congestion

TCP Slow Start

- when connection begins, increase rate exponentially until first loss event or threshold is reached, after which **`cwnd`** grows more slowly (linearly - congestion avoidance CA phase):
  - initially **`cwnd`** = 1 MSS
  - double **`cwnd`** every RTT
  - done by incrementing **`cwnd`** for every ACK received
- *summary:* initial rate is slow but ramps up exponentially fast – 1, 2, 4..
- linear growth after threshold



TCP: detecting, reacting to loss

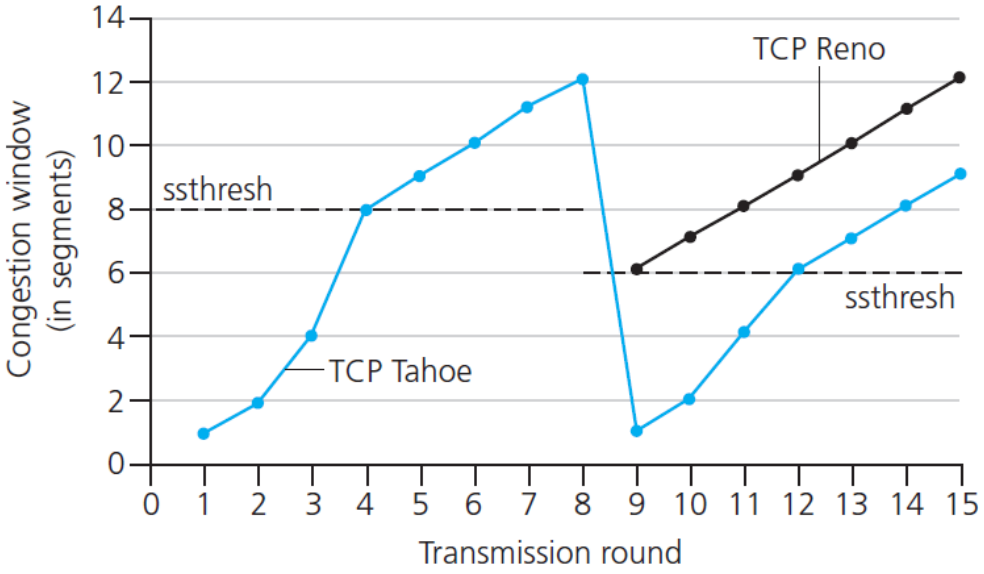
- loss indicated by timeout:
  - **`cwnd`** set to 1 MSS again;
  - window then grows exponentially (as in slow start) to threshold, then grows linearly
- loss indicated by 3 duplicate ACKs (1+3 in total):
  - TCP Tahoe sets **`cwnd`** to 1 MSS and goes to slow start again (same behavior as for timeout)
  - resends packet
- TCP RENO
  - dup ACKs indicate network capable of delivering some segments
  - **`cwnd`** is cut in half window
  - resends packet
  - goes into fast recovery – if it receives ACK it stays in linear mode, else it goes into slow start.

TCP: switching from slow start to CA

Q: when should the exponential increase switch to linear?

A: when **`cwnd`** gets to a threshold.

Note: here we show Reno in fast recovery mode.

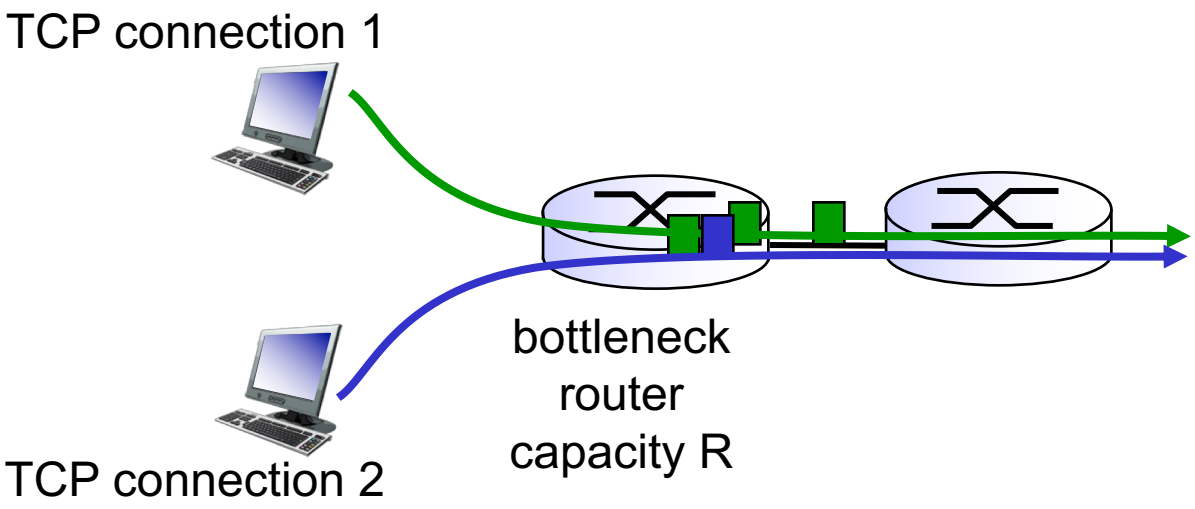


Implementation:

- variable **`ssthresh`**, starts with default value
- on loss event, **`ssthresh`** is set to 1/2 of **`cwnd`** just before loss event

TCP Fairness

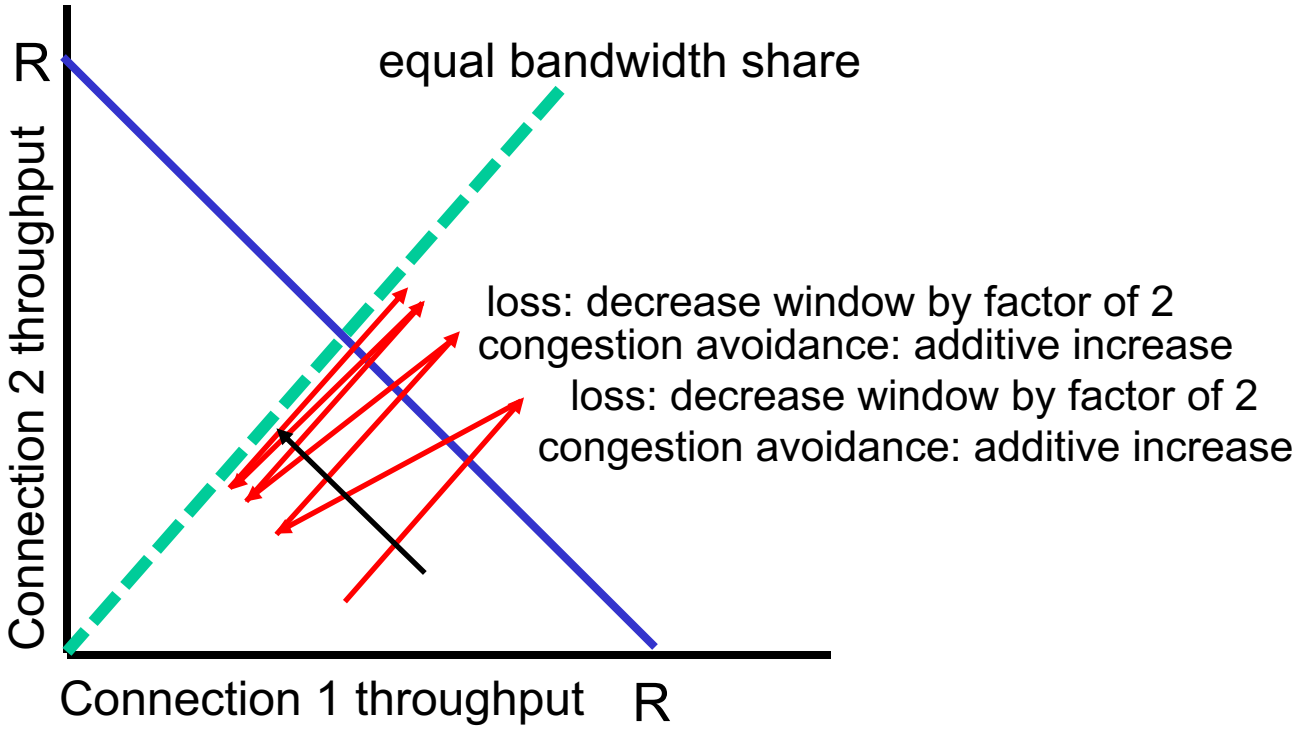
fairness goal: if K TCP sessions share same bottleneck link of bandwidth R, each should have average rate of R/K



Why is TCP fair?

two competing sessions:

- additive increase gives slope of 1, as throughout increases
- multiplicative decrease decreases throughput proportionally



Fairness (more)

Fairness and UDP

- multimedia apps often do not use TCP
  - do not want rate throttled by congestion control
- instead use UDP:
  - send audio/video at constant rate, tolerate packet loss

Fairness, parallel TCP connections

- application can open multiple parallel connections between two hosts
- web browsers do this
- e.g., link of rate R with 9 existing connections:
  - new app asks for 1 TCP, gets rate R/10 (1/(9+1=10)R)
  - new app asks for 11 TCPs, gets ~ R/2 (11/(9+11=20)R)

Explicit Congestion Notification (ECN)

network-assisted congestion control:

- two end points negotiated ECN capability in setup phase with options in SYN, SYN ACK packets)
- two bits in IP header (ToS field) marked *by network router* to indicate congestion capability (01 or 10) and congestion (11)
- congestion indication carried to receiving host (packet not dropped)
- receiver (seeing 11 in IP datagram) sets ECE bit on receiver-to-sender ACK segment to notify sender of congestion (unused flag fields in TCP)
- sender reduces cwnd and sets CRW bit (second unused flag bit) on next segment transmission